

Faeze Mohammadi

[GitHub](#)[LinkedIn](#)[Email](#)[Kaggle](#)

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SUMMARY

Computer Engineering student at the University of Tehran with interests across artificial intelligence, machine learning, computer vision, algorithms, and computer systems. Built projects that combine implementation with evaluation and analysis; gained research-writing experience through a Higgs boson review article. Interested in collaborative research grounded in careful reasoning, transparent experiments, and clear communication.

EDUCATION

University of Tehran - Computer Engineering undergraduate

Selected course grades (20/20 in each): Probability and Statistics; Discrete Mathematics; Advanced Programming; Artificial Intelligence; Data Structures.

RESEARCH AND ACADEMIC WRITING

A Study of the Higgs Boson Particle

Sole author | Review article | Manuscript available upon request

- Reviewed and synthesized scientific literature on the Higgs boson.
- Organized and wrote the manuscript to present technical ideas clearly in a structured, research-oriented format.

TEACHING EXPERIENCE

Teaching Assistant - Formal Languages and Automata

University of Tehran | Instructor: Dr. Hossein Hojjat

- Supported students with formal models, regular languages, grammars, finite automata, and course problem solving.

Teaching Assistant - Discrete Mathematics

University of Tehran | Instructor: Dr. Siamak Mohammadi

- Helped students work through logic, proofs, relations, combinatorics, graphs, and discrete structures.

SELECTED PROJECTS

Blood Cell Classification with PyTorch

[GitHub](#)

Compared multilayer perceptron and convolutional network models for blood-cell classification. Examined feature maps, errors, spatial robustness, and occlusion sensitivity to better understand model behavior.

Methods / tools: PyTorch; computer vision; error and robustness analysis.

Supervised Learning from Scratch

[GitHub](#)

Implemented decision trees and softmax regression, compared them with scikit-learn baselines, and investigated feature engineering and class imbalance.

Methods / tools: Python; supervised learning; model evaluation.

Genetic Music and Minimax Checkers

[GitHub](#)

Explored music reconstruction with genetic algorithms and adversarial checkers search with Minimax and Alpha-Beta pruning; documented experiments and visual results.

Methods / tools: Optimization; adversarial search; experimental comparison.

PROJECTS CONTINUED

Snake Search with Portals

[GitHub](#)

Applied BFS, DFS, IDS, A*, and Weighted A* to a portal-based pathfinding problem. Compared how the search methods behave in an interactive setting.

Methods / tools: Graph search; heuristic search; visualization.

Pipelined RISC-V CPU

[GitHub](#)

Built a five-stage RV32I processor with forwarding, hazard detection, and branch handling. Used program-level tests to verify the pipeline.

Methods / tools: Verilog; computer architecture; verification.

ESP8266 Smart Irrigation

[GitHub](#)

Created a soil-moisture monitoring system with automatic pump control and a local web dashboard for observing the device.

Methods / tools: ESP8266; embedded systems; local web interface.

Further statistical work: [Bayesian robot localization in R](#) with noisy sensing and motion, and [Bernoulli Naive Bayes in R](#) for probabilistic spam classification.

COURSES AND CERTIFICATES

Coursera - DeepLearning.AI / Stanford Online

Machine Learning Specialization

[PDF](#)

Supervised Machine Learning: Regression and Classification

[PDF](#)

Advanced Learning Algorithms

[PDF](#)

Unsupervised Learning, Recommenders, Reinforcement Learning

[PDF](#)

Introduction to Statistics

[PDF](#)

Writing in the Sciences

[PDF](#)

OpenCV University

Free Vision Language Models (VLM) Bootcamp

[PDF](#)

Free TensorFlow-Keras Bootcamp

[PDF](#)

Free PyTorch Bootcamp

[PDF](#)

Python for Beginners

[PDF](#)

Free OpenCV Bootcamp

[PDF](#)

TECHNICAL TOOLKIT

Programming and hardware: Python, C++, C, R, Verilog, SystemVerilog, HTML/CSS.

AI, ML, and data: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, NumPy, pandas, Jupyter Notebook.

Systems and tools: RISC-V, FPGA, ModelSim, Quartus, LTspice, Arduino, ESP8266.

Foundations: Algorithms, Data Structures, Probability, Statistics, Formal Languages and Automata, Computer Architecture.

WORKING APPROACH

I prefer projects where the reasoning can be inspected: experiments, verification results, visualizations, and comparisons sit alongside the implementation. I am strengthening my foundations in AI and ML while exploring connections with reliable software and hardware-aware computing.